

PROJECT 16

Configure S3 Lifecycle Expiration for CloudTrail Log Cost Control

Amazon S3 | Lifecycle Policies | CloudTrail | LogArchive

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Signed in through **IAM Identity Center**, assumed administrative access into the **LogArchive** account, and created an Amazon S3 lifecycle rule named **Delete old CloudTrail**. The rule applies to the entire CloudTrail log bucket and expires current object versions after **90 days** to control storage cost in the lab environment.

COMPLIANCE
NOTE

This configuration is intentionally documented as a lab cost-saving control. It is not presented as a standard production log-retention policy because production environments may require longer retention, immutable storage, Object Lock, Glacier archive, or regulatory retention controls.

Business Problem

Organization CloudTrail logging centralizes audit records in S3, but the log bucket grows over time and creates storage cost. For this lab environment, retaining every CloudTrail object indefinitely is unnecessary. An S3 lifecycle policy provides a controlled way to expire older objects and keep the account cost low while still preserving recent lab activity for review.

Objectives

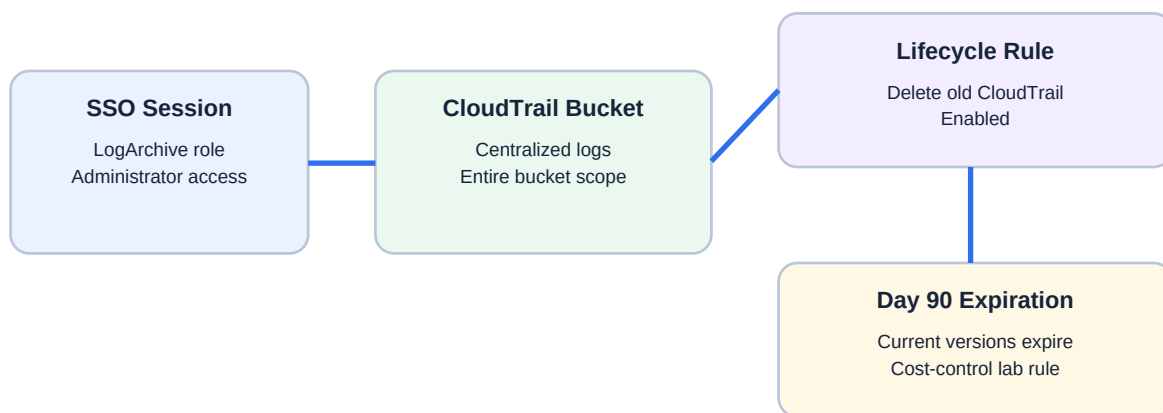
- Use IAM Identity Center to sign into the AWS access portal.
- Assume the LogArchive account administrator role.
- Open the centralized CloudTrail S3 bucket.
- Create an S3 lifecycle rule named Delete old CloudTrail.
- Apply the rule to the entire bucket.
- Expire current versions of objects after 90 days.
- Document the difference between lab cost-saving retention and production compliance retention.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	Amazon S3
Access Path	IAM Identity Center SSO
Account Context	LogArchive
Role	Administrator access / assumed LogArchive role
Bucket Purpose	Centralized CloudTrail log storage

Lifecycle Rule Name	Delete old CloudTrail
Lifecycle Rule Status	Enabled
Scope	Entire bucket
Current Version Action	Objects uploaded on Day 0; objects expire on Day 90
Noncurrent Version Action	No actions defined
Purpose	Lab cost control for CloudTrail S3 storage

Architecture



The lifecycle policy expires current CloudTrail log objects after 90 days to control storage cost in the lab.
This is intentionally documented as a lab cost-saving choice, not a production compliance-retention baseline.

Figure 1 - SSO access into LogArchive and lifecycle expiration on the CloudTrail bucket.

Implementation Summary

1. Signed into the IAM Identity Center access portal.
2. Selected the LogArchive account and assumed administrator access.
3. Opened Amazon S3 in the LogArchive account context.
4. Selected the centralized CloudTrail log bucket.
5. Opened the bucket Management tab.
6. Created a new lifecycle rule named Delete old CloudTrail.
7. Applied the lifecycle rule to all objects in the bucket.

- 8. Selected Expire current versions of objects.
- 9. Set expiration to 90 days after object creation.
- 10. Saved the rule and confirmed that it is enabled.

Lifecycle Rule Behavior

Rule name	Delete old CloudTrail	Identifies the lifecycle control for CloudTrail log cleanup.
Status	Enabled	The rule is active.
Scope	Entire bucket	Applies to all objects in the CloudTrail bucket.
Current object expiration	90 days	Current object versions expire 90 days after creation.
Noncurrent version actions	None	No separate noncurrent version cleanup is configured in the evidence.
Delete markers / incomplete uploads	None	No explicit cleanup action for expired delete markers or incomplete multipart uploads is shown.

Evidence

Figure 2 - Enabled S3 Lifecycle Rule for CloudTrail Log Expiration

Delete old CloudTrail info

EditDeleteActions

Lifecycle rule configuration

Lifecycle rule name

Delete old CloudTrail

Status

Enabled

Scope

Entire bucket

Prefix

-

Object tags

-

Minimum object size

-

When no minimum object size is specified, the minimum object size for transitions is determined by the lifecycle configuration. [Learn more](#)

Maximum object size

-

Review transition and expiration actions

Current version actions

Day 0

Objects uploaded

↓

Day 90

Objects expire

Noncurrent versions actions

Day 0

No actions defined.

Delete expired object delete markers or incomplete multipart uploads

Expired object delete markers

-

Incomplete multipart uploads

-

The S3 lifecycle rule evidence shows the rule named Delete old CloudTrail with Enabled status, scope set to the entire bucket, and current version action configured so objects uploaded on Day 0 expire on Day 90. The noncurrent version action area shows no actions defined.

Security and Compliance Analysis

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The lifecycle rule is appropriate for a cost-conscious lab because it limits storage growth while preserving recent CloudTrail logs for short-term review.

This is not a production retention baseline. Production CloudTrail logs often need longer retention based on incident response, audit, legal, regulatory, or business requirements.

Deleting logs too quickly can reduce forensic visibility. The 90-day setting should be treated as a deliberate lab tradeoff rather than a general recommendation.

For production, S3 versioning, Object Lock, Glacier archive, lifecycle transition rules, and SCP or IAM controls that prevent destructive actions may be required.

Because the evidence shows no noncurrent version action, this project should not be represented as a complete versioned-object retention strategy.

Lessons Learned

S3 lifecycle rules can manage object retention automatically by expiring or transitioning objects based on age.

CloudTrail service use may be free for the first trail, but S3 storage for delivered logs still creates cost.

Cost optimization and compliance retention can conflict. The correct policy depends on the environment's purpose and requirements.

Lifecycle rules can be useful for security data management, but they must be paired with retention, integrity, and recovery requirements when used in production.

Production Improvements

- Define CloudTrail retention based on audit, legal, and incident-response requirements before deleting logs.
- Use S3 lifecycle transitions to lower-cost storage classes instead of deleting logs when retention is required.
- Enable S3 versioning and evaluate Object Lock for immutable log retention.
- Use Glacier archive or Glacier Deep Archive where long-term retention is needed at lower cost.
- Monitor lifecycle configuration changes through CloudTrail and centralized alerts.
- Document who is authorized to change log-retention and lifecycle policies.

Skills Demonstrated

S3 lifecycle policies	CloudTrail retention	IAM Identity Center
Object expiration	Lab cost optimization	LogArchive access
Retention analysis	Compliance caveats	Centralized logging operations

Resume Bullet

RESUME

Configured an Amazon S3 lifecycle policy for a centralized CloudTrail log bucket by using IAM Identity Center to access the LogArchive account and enabling a 90-day object-expiration rule for lab cost control while documenting production retention caveats.

STAR Interview Story

Situation: The AWS lab environment was collecting organization CloudTrail logs in S3, creating ongoing storage cost.

Task: Reduce lab storage cost without presenting the setting as a production compliance-retention standard.

Action: I signed in through IAM Identity Center, assumed the LogArchive administrator role, opened the CloudTrail bucket, created the Delete old CloudTrail lifecycle rule, applied it to the entire bucket, and set current objects to expire after 90 days.

Result: The lab now has an enabled lifecycle policy that automatically expires old CloudTrail objects after 90 days, reducing storage cost while keeping the compliance limitation clearly documented.

Version History

v1.0.0	July 13, 2026	Initial documented implementation. Signed into SSO, assumed the LogArchive role, created an enabled S3 lifecycle rule named Delete old CloudTrail, configured current object expiration after 90 days, documented lab cost-saving scope and production compliance caveats, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - On the Meaning of Life(cycles), Versions, and Ransomware.
- User-provided S3 lifecycle rule evidence.
- Project 12 - Organization CloudTrail Centralized Logging.
- Project 15 - AWS Organizations Enterprise OU Structure.